

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279649

Kind Code

A1

Publication Date

September 04, 2025

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SYSTEMS AND METHODS FOR WORKSITE DYNAMIC CHARGING

Abstract

At least one aspect of the present disclosure is directed to a system, method, or a computer-readable medium of the following. A method can include one or more processors receiving information indicative of load metrics relating to a power grid. The method can include the one or more processors identifying metrics for a plurality of charging machines to be charged at the worksite. The method can include the one or more processors determining a charging rate for a charging machine at the worksite, for each charging machine of the plurality of charging machines, according to the metrics and load metrics of the power grid. The method can include the one or more processors controlling a plurality of chargers at the worksite, according to the charging rate associated with a corresponding charging machine.

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Appl. No.: 18/592761

Filed: March 01, 2024

Publication Classification

Int. Cl.: H02J3/16 (20060101); H02J3/32 (20060101); H02J3/38 (20060101); H02J7/00 (20060101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present implementations relate generally to dynamic charging, and more particularly to worksite dynamic charging.

INTRODUCTION

[0002] The present disclosure relates generally to dynamic charging, and more particularly to worksite dynamic charging. Worksites, such as construction sites, manufacturing plants, mining sites, and/or agricultural sites, may require machines to charge various components/hardware/vehicles on or at the worksite.

SUMMARY

[0003] This disclosure is generally directed to worksite dynamic charging. With the introduction of electrified machines in a worksite, there is an increase of vehicle to vehicle charging and vehicle to grid charging. The process of vehicle to vehicle charging and vehicle to grid charging at a worksite can raise issues of compatibility, because of the different charging requirements of different machines. The process can reduce productivity, increase energy demand, raise scheduling issues, increase project timelines and increase cost to maintain a higher charging infrastructure. Aspects of this technical solution can provide a method for worksite dynamic charging to improve productivity, reduce compatibility issues, reduce energy demand, reduce scheduling issues, and reduce the cost of the charging infrastructure. For example, using worksite dynamic charging allows for a method that can include receiving information indicative of load metrics relating to a power grid. The method can include identifying metrics for a plurality of charging machines to be charged at a worksite. The method can include determining, for each charging machine of the plurality of charging machines, a charging rate for the charging machine at the worksite, according to the metrics and the load metrics of the power grid. The method can include controlling a plurality of chargers at the worksite, according to the charging rate associated with a corresponding charging machine.

[0004] A first aspect provided herein relate to a method of dynamic charging at a worksite. The method can include one or more processors receiving information indicative of load metrics relating to a power grid. The method can include the one or more processors identifying metrics for a plurality of charging machines to be charged at the worksite. The method can include the one or more processors determining a charging rate for a charging machine at the worksite, for each charging machine of the plurality of charging machines, according to the metrics and load metrics of the power grid. The method can include the one or more processors controlling a plurality of chargers at the worksite, according to the charging rate associated with a corresponding charging machine.

[0005] In some embodiments, the information indicative of the load metrics is received from the plurality of chargers and from a grid controller associated with the power grid. In some embodiments, the load metrics can include one or more measurements corresponding to a demand in power at the worksite. In some embodiments, the demand can include an increase in power at the worksite or a decrease in power at the worksite.

[0006] In some embodiments, the charging rate for the charging machine can include a negative charging rate. In some embodiments, the method can include controlling the charger according to the negative charging rate can include discharging the corresponding charging machine to the power grid. In some embodiments, the power grid can include a microgrid. In some embodiments,

the one or more processors are of a worksite controller. In some embodiments, the method can include transmitting, by the one or more processors of the worksite controller, one or more signals to a charge/discharge controller, to cause the charge/discharge controller to transmit one or more signals to the plurality of charging machines.

[0007] In some embodiments, the method can include transmitting, by the charge/discharge controller, one or more signals to the plurality of chargers. In some embodiments, the one or more signals can include to charge the charging machine or discharge the charging machine. In some embodiments, the method can include determining, by the worksite controller, the charging rate for the charging machine at the worksite, based on a renewable availability from the microgrid. In some embodiments, the renewable availability including at least one of wind turbines, photovoltaic (PV) panels, generator sets, or hydropower and determining, by the worksite controller, the charging rate for the charging machine at the worksite, based on an efficiency metric for the microgrid to maximize the efficiency of the microgrid.

[0008] In some embodiments, the method can include determining a charging rate for the charging machine at the worksite further comprising receiving, by the one or more processors, battery metrics of the charging machine. In some embodiments, the battery metrics can include at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters. In some embodiments, the method can include receiving, by the one or more processors, a schedule. In some embodiments, the schedule can include one or more gaps in operation of the charging machines.

[0009] A first aspect provided herein relate to a system of dynamic charging at a worksite. The system can include memory and one or more processors that can receive information indicative of load metrics relating to a power grid. The one or more processors can identify metrics for a plurality of charging machines to be charged at a worksite. The one or more processors can determine for each charging machine of the plurality of charging machines, a charging rate for the charging machine at the worksite, according to the metrics and the load metrics of the power grid. The one or more processors can control a plurality of chargers at the worksite, according to the charging rate associated with a corresponding charging machine.

[0010] In some embodiments, the information indicative of the load metrics is received from the plurality of chargers and from a grid controller associated with the power grid. In some embodiments, the load metrics can include one or more measurements corresponding to a demand in power at the worksite. In some embodiments, the demand can include an increase in power at the worksite or a decrease in power at the worksite.

[0011] In some embodiments, the charging rate for the charging machine can include a negative charging rate. In some embodiments, to control the charger according to the negative charging rate, the one or more processors are configured to discharge the corresponding charging machine to the power grid. In some embodiments, the metrics for the plurality of charging machines can include one or more measurements corresponding to a supply of charge with the corresponding charging machine.

[0012] In some embodiments, the power grid can include a microgrid. In some embodiments, the one or more processors are of a worksite controller. In some embodiments, the worksite controller is configured to transmit one or more signals to a charge/discharge controller, to cause the charge/discharge controller to transmit one or more signals to the plurality of charging machines. In some embodiments, the charge/discharge controller is configured to transmit one or more signals to the plurality of chargers. In some embodiments, the one or more signals can include charge the charging machine or discharge the charging machine.

[0013] In some embodiments, the worksite controller is configured to determine the charging rate for the charging machine at the worksite, based on a renewable resource availability from the microgrid. In some embodiments, the renewable resource availability can include at least one of wind turbines, photovoltaic (PV) panels, generator sets, or hydropower determine the charging rate

for the charging machine at the worksite, based on an efficiency metric for the microgrid to maximize efficiency of the microgrid.

[0014] In some embodiments, the one or more processors are configured to receive battery metrics of the charging machine. In some embodiments, the battery metrics indicate at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters and receive a schedule. In some embodiments, the schedule indicates one or more gaps in operation of the charging machines. In some embodiments, the gap is a period of time in which the charging machine is not in use at the worksite.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] These and other aspects and features of the present implementations will become apparent to those ordinarily skilled in the art upon review of the following description of specific implementations in conjunction with the accompanying figures.

[0016] FIG. 1 depicts an example system to implement worksite dynamic charging at a worksite, in accordance with present implementations.

[0017] FIG. 2 depicts example sections of the worksite and a microgrid, in accordance with present implementations.

[0018] FIG. 3 depicts an example layout of the worksite, in accordance with present implementations.

[0019] FIG. 4 depicts an example data processing system for to implement worksite dynamic charging, in accordance with present implementations.

[0020] FIG. 5 depicts an example flow diagram for power quality support for the worksite and a microgrid, in accordance with present implementations.

[0021] FIG. 6 depicts an example flow diagram for optimization of charging for the worksite and the microgrid, in accordance with present implementations.

[0022] FIG. 7 depicts a method to implement worksite dynamic charging for the worksite and microgrid, in accordance with present implementations.

DETAILED DESCRIPTION

[0023] This disclosure is generally directed to a method for worksite dynamic charging to increase efficiency while reducing costs at a worksite. With the introduction of electrified machines in a worksite, there is an increase of vehicle to vehicle charging and vehicle to grid charging. The process of vehicle to vehicle charging and vehicle to grid charging at a worksite can raise issues of compatibility, because of the different charging requirements of different machines. The process can reduce productivity, increase energy demand, raise scheduling issues, increase project timelines and increase cost to maintain a higher charging infrastructure at the worksite. Aspects of this technical solution can provide a method for worksite dynamic charging to improve productivity, reduce compatibility issues, reduce energy demand, reduce scheduling issues, and reduce the cost of the charging infrastructure. For example, using worksite dynamic charging allows for a method that can include receiving information indicative of load metrics relating to a power grid. The method can include identifying metrics for a plurality of charging machines to be charged at a worksite. The method can include determining for each charging machine of the plurality of charging machines, a charging rate for the charging machine at the worksite, according to the metrics and the load metrics of the power grid. The method can include controlling a plurality of chargers at the worksite, according to the charging rate associated with a corresponding charging machine.

[0024] FIG. 1 depicts an example system **100** to implement worksite dynamic charging at a worksite. The system **100** can include a data processing system **102**, operation locations **104A-N**

(referred to as operation locations **104** herein), and chargers **106A-N** (referred to as chargers **106** herein). The above-mentioned components may be connected to each other through a network **101**. The examples of the network **101** may include, but are not limited to, cellular (e.g., 3G, 4G, LTE, 5G, etc.) network, private or public local area network (LAN), wireless-LAN (WLAN), metropolitan area network (MAN), wide area network (WAN), and so forth, which may be used for communicating (e.g., via the Internet) with various endpoints. The network **101** may include both wired and wireless communications according to one or more standards and/or via one or more transport mediums.

[0025] The communication over the network **101** may be performed in accordance with various communication protocols such as Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), and IEEE communication protocols. In one example, the network **101** may include wireless communications according to Bluetooth specification sets, or another standard or proprietary wireless communication protocol. In another example, the network **101** may also include communications over a cellular network, including, e.g., a GSM (Global System for Mobile Communications), CDMA (Code Division Multiple Access), EDGE (Enhanced Data for Global Evolution) network.

[0026] The system **100** is not confined to the components described herein and may include additional or alternate components, not shown for brevity, which are to be considered within the scope of the embodiments described herein.

[0027] The system **100** can include at least one data processing system **102**. The data processing system **102** can include a physical computer system operatively coupled or that can be coupled with one or more components of the system **100**, either directly or through an intermediate computing device or system. The data processing system **102** can include a virtual computing system, an operating system, and/or a communication bus to effect communication and processing. The data processing system **102** can include a microgrid controller **110**, a worksite controller **112**, a system processor **114**, a charge/discharge controller **116**, and memory **118**.

[0028] The system **100** can include at least memory **118** within the data processing system **102**. The memory **118** can be coupled to the microgrid controller **110**, the worksite controller **112**, the system processor **114**, and the charge/discharge controller **116**. The memory **118** can store data associated with the system **100**. The memory **118** can include one or more hardware memory devices to store binary data, digital data, or the like. The memory **118** can include one or more electrical components, electronic components, programmable electronic components, reprogrammable electronic components, integrated circuits, semiconductor devices, flip flops, arithmetic units, or the like. The memory **118** can include at least one of a non-volatile memory device, a solid-state memory device, a flash memory device, and a NAND memory device. The memory **118** can include one or more addressable memory regions disposed on one or more physical memory arrays. A physical memory array can include a NAND gate array disposed on, for example, at least one of a particular semiconductor device, integrated circuit device, or printed circuit board device.

[0029] The microgrid controller **110**, the worksite controller **112**, the system processor **114**, and the charge/discharge controller **116** (referred to as the controllers) can execute one or more instructions associated with the data processing system **102**. The controllers can include an electronic processor, an integrated circuit, or the like including one or more of digital logic, analog logic, digital sensors, analog sensors, communication buses, volatile memory, nonvolatile memory, and the like. The controllers can include, but are not limited to, at least one microcontroller unit (MCU), microprocessor unit (MPU), central processing unit (CPU), graphics processing unit (GPU), physics processing unit (PPU), embedded controller (EC), or the like. The controllers can include memory operable to store or storing one or more instructions for operating components of the processors and operating components operably coupled to the controllers. For example, the one or more instructions can include one or more of firmware, software, hardware, operating systems, embedded operating systems. The controllers or the data processing system **102** generally can

include one or more communication bus controllers to effect communication between the controllers and the other elements of the data processing system **102**.

[0030] Referring to FIG. 1 and FIG. 2, FIG. 2 depicts example sections **200** of the worksite **210** and a microgrid **202**. The worksite controller **112** can be configured to receive information indicative of load metrics relating to a power grid (or utilities **204**). The power grid can be an electrical grid or microgrid, with a plurality of interconnected electrical components (e.g., wires, capacitors, inductors, resistors, transformers) to facilitate the generation and distribution of electricity. For example, the power grid may be electrically connected to a power plant that generates electricity, to distribute the power to various service drops or endpoints. Using one or more transmission lines, the power grid can distribute the electricity to one or more chargers **106**. The information indicative of load metrics can include electricity generation data, load data, voltage data, transmission line data, fault data, or weather data. The information can be used to form the load metrics. For example, a worksite controller **112** can receive electricity generation data. In another example, a worksite controller **112** can receive load data and fault data. In yet another example, a worksite controller **112** can receive transmission line data, electricity generation data, and fault data.

[0031] The system **100** can include at least one charger **106** at a worksite **210**. The worksite can include one or more operation locations **104**. The operation location **104** can be a specific area within the worksite **210** where a plurality of activities corresponding to the work environment take place. For example, at a construction site, an operation location **104** can be an excavation site or a building assembly site. In another example, at mining site, an operation location **104** can be an area or location where mineral/material/fluids/ore extraction occurs. Each operation location **104** can include one or more machines **108A-N** (referred to as machines **108** herein). For example, referring briefly to FIG. 3, operation location **104A** can include machine **108A**, operation location **104B** can include machine **108B**, and the worksite **300** can generally include chargers **106A-106C** and machines **108C-108G** displaced or positioned throughout the worksite **300**.

[0032] Referring still to FIG. 1 and FIG. 2, the section **200** may include a microgrid **202**. The microgrid **202** may be or include a localized or contained power grid dedicated for a particular region. The microgrid **202** may include various power sources **203** which supply power to the microgrid for distribution in the localized/contained area. For example, the microgrid **202** may include power sources **203**, including utilities **204**, generator sets **206**, and renewable power sources **208**. Various combinations of such power sources **203** can supply power to the microgrid **202**, to supply power to the localized/contained area (such as the worksite **210**).

[0033] The power sources **203** which supply power to the microgrid **202** may include utilities **204**. The utilities **204** may be or include a power source or connector corresponding to a utility line (e.g., a line/service drop from the power grid) supplying power generated for distribution across a wide power grid. The power of the utilities can be supplied by a power plant (e.g., solar power plant, hydroelectric power plant, or wind power plant, among others). The power plants can be owned or operated by an energy company, to supply power to the worksite **210**. For example, the power plant can supply power through a utility line of the utilities **204**, and the utilities **204** can provide the power to the microgrid **202**.

[0034] The power sources **203** which supply power to the microgrid **202** may include generator sets **206** (referred to as “gensets **206**” herein). The gensets **206** may be located within close proximity to the operation locations **104**, chargers **106**, and/or areas of high power consumption. The gensets **206** may include backup generators, distributed energy resources (DERs), or portable generators among others. For example, as shown in FIG. 3, a genset **206** may be located next to an operation location **104A** to supply power directly to the machines **108** at the operation location **104A**. In another example, the gensets **206** may be located next to one or more chargers **106** to supply power directly to the one or more chargers **106**.

[0035] The power sources **203** which supply power to the microgrid **202** may include renewable

energy sources (referred to as “renewables **208**” herein). The renewables **208** can include solar power, wind power, hydropower, geothermal power, wave energy, or biomass energy, communicably coupled to a power store configured to store power sourced from such renewable energy sources. The renewables **208** can naturally generate power over time and are sustainable, and have a lower environmental impact. The renewables **208** can receive energy from solar farms, wind turbines, water dams, geothermal power plants, or wave energy converters, among others. For example, a solar farm can supply power in the form of solar energy to the microgrid **202**. In another example, wind turbines can supply power in the form of wind energy to the microgrid **202**. [0036] The worksite **210** can include one or more buildings **212**. The buildings **212** may be associated with the worksite **210**. For example, an excavation worksite **210** can include one or more control buildings **212** to monitor and control complex processes related to production and energy consumption. The worksite **210** can include one or more industrial sites **214**. The industrial sites **214** can be a specific area of the worksite **210** relating to manufacturing, production, or processing at the worksite **210**. For example, an industrial site **214** can be a manufacturing plant at a worksite **210**. In another example, an industrial site **214** can be a refinery. The worksite **210** can include a section for charging **216** (also referred to as “charging sites” **216**). The charging sites **216** can include the chargers **106** and the machines **108**.

[0037] The chargers **106** can be a device or a system that supplies electrical power for the charging of the battery of the machines **108** at the worksite. For example, a machine **108A** can connect to a charger **106A** to replenish the energy stored in the batteries of the machine **108A**. Referring to FIG. 1 and FIG. 3, FIG. 3 depicts an example layout **300** of the worksite **210**. The chargers **106** can be at a plurality of locations at the worksite. The chargers **106** can be near operation locations **104** or near machines **108**. For example, a charger **106A** may be located at or near operation location **104B**, where the operation location **104B** includes machine **104B**. In another example, a charger **106B** may be located near machines **108E** and **108F**, respectively. The chargers **106** can be or include a portable generator to move toward machines **108** which cannot reach the chargers **106**. For example, a charger **106A** can move towards a machine **108B** (e.g., at operation location **104B**) in the case where a battery pack within the machine **108A** has depleted all of the stored energy.

[0038] The machines **108** can be at least one of charging machines, heavy equipment, electric machines, or heavy machines. Each machine **108** in the machines **108** can be designed to fulfill tasks corresponding to the operation locations **104** at the worksite. For example, at an excavation site the machines **108** can include bulldozers, excavators, or drill rigs, among others. Each machine **108** can include a transmission system, an engine system, a safety system, a communications system, or a hydraulic system among others. The machines **108** can use the communication system to transmit data and receive data or instructions to and from with the data processing system **102** over the network **101**, respectively. For example, a machine **108** can send battery information and location information to a worksite controller **112**. In another example, a charge/discharge controller **116** can transmit a charge command to a machine **108**.

[0039] Referring back to FIG. 1 and FIG. 2, the worksite controller **112** can receive information indicative of the load metrics from the chargers **106**. The chargers **106** can transmit the information over the network **101**. In some embodiments, the chargers **106** may receive a request from the worksite controller **112**. The request can trigger the chargers **106** to transmit the information to the worksite controller **112**. For example, a worksite controller **112** can transmit a request to one or more chargers **106** and cause the one or more chargers **106** to transmit information to the worksite controller **112**. The information can include active power of the chargers **106**, reactive power of the chargers **106**, voltage of the chargers **106**, and the frequency associated with the chargers **106**. The active power of the chargers **106** can correspond to power consumed by the chargers **106** from the electrical supply and convert the power to an appropriate form corresponding to the machines **108** at the chargers **106**. The reactive power of the chargers **106** can correspond to electrical power which oscillates between a source and a load withing being consumed as active power. In some

embodiments, the chargers **106** can periodically transmit such information to the worksite controller **112**. In this regard, and in various embodiments, the chargers **106** can periodically, aperiodically, and/or on-demand transmit such information to the worksite controller **112**.

[0040] The worksite controller **112** can receive information indicative of the load metrics from the microgrid controller **110**. The microgrid controller **110** can transmit the information over the network **101**. In some embodiments, similar to the chargers **106**, the microgrid controller **110** may receive a request from the worksite controller **112**. The request can trigger the microgrid controller **110** to transmit the information to the worksite controller **112**. For example, a worksite controller **112** can transmit a request to a microgrid controller **110** and cause the microgrid controller **110** to transmit information to the worksite controller **112**. The information from the microgrid controller **110** can provide insights regarding the demand and consumptions of electrical energy at the microgrid. In this regard, the microgrid controller **110** may provide such information on-demand (e.g., in response to a request from the worksite controller **112**). In some embodiments, the microgrid controller **110** can periodically transmit such information to the worksite controller **112**. In this regard, and in various embodiments, the microgrid controller **110** can periodically, aperiodically, and/or on-demand transmit such information to the worksite controller **112**.

[0041] The load metrics can include one or more measurements corresponding to a demand in power at the worksite. The demand can indicate an increase or decrease in demand at the worksite. For example, a worksite may have a high demand. The high demand indicates there is a significant increase in power consumption at the worksite. In another example, a worksite may have a low demand. The low demand indicates that is a significant decrease in power consumption at the worksite. The increase in power can correspond to the use of the machines **108**. For example, a first machine **108A**, a second machine **108B**, a third machine **108C**, a fourth machine **108D**, and a fifth machine **108E** perform various activities (e.g., power consuming activities) at one or more locations at the worksite. Since there are multiple machines **108** active, the demand can increase during a period of time in which any number of such machines are actively charging (e.g., via the corresponding chargers **106**). The one or more measurements can associate with the machines **108** during an activity associated with the worksite. The one or more measurements can include machine **108** utilization, machine **108** downtime, energy consumption, fuel consumption, worksite traffic, task completion time, and equipment health monitoring, among others. For example, a load metric can be a combination of machine **108** utilization, energy consumption of the machine **108**, and machine **108** downtime. The load metrics can be stored in memory **118** for reference during a future time period.

[0042] The worksite controller **112** can identify metrics for a plurality of charging machines **108** to be charged at the worksite. The metrics can include one or more measurements corresponding to a supply of charge with the corresponding machine **108**. The supply of charge can indicate the stored energy with a battery of the machine **108**. For example, a machine **108A** can have a supply of charge of 20%. In another example, a machine **108B** can have a supply of charge of 90%. The one or more measurements can include energy consumption, operating hours, idle time, productivity, temperature metrics, and operator behavior, among others. For example, a metric can be a combination of energy consumption and operating hours for a machine **108A**. In another example, the metric can be a combination of productivity, idle time, and energy consumption for a machine **108A**.

[0043] The worksite controller **112** can receive battery metrics of the machines **108** from the machines **108** and/or from the chargers **106**. For example, a machine **108A** can transmit battery metrics to the worksite controller **112** via a network **101**. In another example, a charger **106A** can transmit battery metrics of a machine **108A** at the charger **106A** (e.g., the machine **108A** connected to and being charged by the charger **106A**) to the worksite controller **112**. The battery metrics can include a state of charge, a state of health, a voltage, charging time, or safety parameters, among others. For example, battery metrics of a machine **108A** can indicate the machine **108A** has a state

of charge of 80%. In another example, battery metrics of a machine **108A** can indicate the machine **108A** has a state of health of 3%. In yet another example, battery metrics of a machine **108A** connected to a charger **106A** can indicate the machine **108A** has a charging time of 3 hours. [0044] The worksite controller **112** can determine, for the machines **108**, a charging rate for the machines **108** at the worksite. The charging rate can be determined for each machine **108** of the machines **108** (e.g., being currently charged or to be charged). For example, machine **108A** can be a bulldozer with a charging rate of 100 A/h, whereas machine **108B** can be an excavator with a charging rate of 85 A/h. Each machine **108** can have different battery characteristics. The battery characteristics of the machines **108** can include battery capacity, battery voltage, battery energy density, battery power density, or battery discharge rate, among others. For example, a machine **108A** can have a battery capacity of 300 amp-hours (AH), whereas a machine **108B** can have a battery capacity of 600 AH.

[0045] The charging rate can be based on the metrics at the worksite **210** and the load metrics of the microgrid **202**. The system processor **114** may execute one or more functions to leverage aspects of the metrics and the load metrics to determine the charging rate. For example, a system processor **114** can use one or more functions to combine aspects of metrics and load metrics. The functions can extract energy consumption of the machines **108** from the metric and energy consumption of the microgrid **202** from the load metric.

[0046] Referring to FIG. 1 and FIG. 4, FIG. 4 depicts an example **400** data processing system **102** for to implement worksite dynamic charging. The system processor **114** can execute the one or more functions using a load calculator **406** based on a signal from the worksite controller **112**. The load calculator **406** can include and analyze and compute metrics and rates related to performance, efficiency, health, and charge for the machines **108**. For example, the worksite controller **112** can send a signal including an identified metric for a plurality of charging machines **108A-N** and a load metric from received information relating to the power source(s) **203** to the system processor **114**. The system processor **114** can use the load calculator **406** to determine a charging rate for one or more machines **108**. The system processor **114** can transmit the charging rate to the worksite controller **112**. For example, after a load calculator **406** calculates a charging rate, a system processor **114** transmit the charging rate to a worksite controller **112**.

[0047] The system processor **114** can include a load predictor **402**. The load predictor **402** can generate one or more predictions for a load of the microgrid **202**. For example, a load predictor **402** can use a load of a microgrid **202** from a previous time period or window (e.g., based on stored/maintained/accessed historic data) to predict the load of the microgrid **202** of a future time period. For instance, the load predictor **402** can identify a previous time period or window based on the current time, and identify or predict the load of the microgrid **202** based on or according to the historic load at the previous time period or window (along with any changes or delta accounted for under current operating conditions). The load predictor **402** can transmit the predicted load to the microgrid controller **110**. For example, a load predictor **402** can send a predicted load for the microgrid **202** to a microgrid controller **110**. The microgrid controller **110** can incorporate the predicted load into the load metrics of the microgrid **202** to determine the charging rate. The microgrid controller **110** may transmit the predicted load to the worksite controller **112**.

[0048] The charging rate for the charging machines **108** can be based on a renewable availability from the microgrid **202**. The renewable availability can correspond to the renewables **208** within the microgrid **202**. The renewable availability can include at least one of wind speed, sunshine, force of tidal waves, or force of water in dams, among others. The renewable availability can correspond to the availability of the renewables **208**. The renewable availability can change at various instances throughout the day. For example, a high renewable availability at the microgrid **202** can indicate a high number of renewables **208** at an instance of a day. The high number of renewables include a high availability of PV panels, gensets **206**, or wind turbines. When the renewable availability is high, the microgrid **202** can rely on energy generated from the renewables

208 to charge the machines **108**.

[0049] The charging rate can be based on an efficiency metric for the microgrid **202** to maximize efficiency of the microgrid **202**. The microgrid controller **110** can generate the efficiency metric based on one or more microgrid assets **410** in memory **118**. The microgrid assets **410** can communicate with the chargers **106** to receive updates for the microgrid assets **410**. For example, a charger **106A** can transmit microgrid assets **410** to the memory **118** throughout a day. The microgrid controller **110** can transmit updates to regarding the microgrid assets **410** to chargers **106**. For example, a microgrid controller **110** can transmit updates to microgrid assets **410** to chargers **106**. The microgrid assets **410** can include energy resources, energy storage, and control systems and the microgrid controller **110** can use elements of the microgrid assets **410** to generate the efficiency metric. For example, an efficiency metric can include data regarding energy resources and energy storage at a microgrid **202**. The efficiency metric can correspond to a highest efficiency point of the microgrid assets **410** at the microgrid **202**. The microgrid controller **110** can observe updates, changes, and data associated with the utilities **204**, gensets **206**, and renewable **208** to develop the microgrid assets **410** and generate the highest efficiency point.

[0050] The solution described herein may solve or implement an optimization function. The optimization function is used to optimize the efficiency metric generated from the microgrid controller **110** by utilizing an efficiency metric of the utilities **204**, an efficiency metric of the gensets **206**, and the efficiency metric of the renewables **208** as inputs for the optimization function. The maximization of the efficiency metric of the utilities **204**, efficiency metric of the gensets **206**, and the efficiency metric of the renewables **208** can directly improve the efficiency metric generated from the microgrid controller **110**. Furthermore, the microgrid controller **110** can provide a recommendation or indication of a charging rate, use of microgrid assets **410**, or power utilization, among others, to the worksite controller **112**. Using the recommendation/indication from the microgrid controller **110**, the worksite controller **112** can vary parameters corresponding to the charge/discharge rate and power quality of the chargers **106** and the machines **108** to ensure the worksite **210** is executing tasks close to the efficiency metric.

[0051] The worksite controller **112** can control the chargers **106** at the worksite **210**, by transmitting one or more signals to the charge/discharge controller **116**. For example, a worksite controller **112** can communicate with a charge/discharge controller **116** via one or more signals over the network **101**. The one or more signals can include instructions, directions, or parameters, to cause the charge/discharge controller **116** to transmit a signal to the chargers **106**. For example, one or more signals from the worksite controller **112** can instruct a charge/discharge controller **116** to adjust one or more parameters of the chargers **106**. The parameters of the charger **106** can include amperage or voltage output, power output, charging mode, charging rate, and/or charge time, among others. For example, a charge/discharge controller **116** can adjust a charging rate of a charger **106A**. In another example, a charge/discharge controller **116** can adjust a charging mode of a charger **106A**. In yet another example, a charge/discharge controller **116** can adjust a power output of a charger **106A**. The signal from the charge/discharge controller **116** can indicate whether to charge the machines **108** or discharge the machines **108**. For example, a signal of a charge/discharge controller **116** indicate that a machine **108A** needs to discharge to a microgrid **202** when at a charger **106A**. In another example, a signal of a charge/discharge controller **116** indicate that a machine **108A** needs to charge at a charger **106A**.

[0052] The worksite controller **112** can control the chargers **106** based on the charging rate associated with a corresponding machine **108**. The charging rate for the machines **108** can include a negative charging rate for the machines **108**. The negative charging rate can indicate a rate that the machines **108** can discharge over time. For example, a machine **108A** can have a negative charging rate by discharging a battery of the machine **108A** during operation or use of the machine **108A**. To control the chargers, the worksite controller **112** can transmit one or more signals to the charge/discharge controller **116**. The one or more signals can indicate to charge the machine **108**.

For example, the worksite controller **112** can transmit a signal to the charge/discharge controller **116**. In response to receiving the signal, the charge/discharge controller **116** can transmit the signal to a machine **108A** to proceed to a charger **106A**. In another example, the worksite controller **112** can transmit a signal to the charge/discharge controller **116**. In response to receiving the signal, the charge/discharge controller **116** can transmit the signal to a machine **108A** to operate, move to, or locate at operation location **104C** until a time window. When the time window occurs the machine **108A** can proceed to the charger **106B**.

[0053] The system processor **114** can include at least one charging scheduler **404**. The charging scheduler **404** can execute one or more instructions associated with the system processor **114**. For example, a system processor **114** can instruct a charging scheduler **404** to generate a schedule for the worksite **210**. The charging scheduler **404** can include one or more communication bus controllers to effect communication between the charging scheduler **404** and the other elements of the data processing system **102**. For example, a charging scheduler **404** can communicate with memory **118**. The charging scheduler **404** can access one or more shift operations **412** in memory **118**. The memory **118** can include the one or more shift operations **412**. The shift operations **412** can be a list, matrix, or data table corresponding to a plurality of tasks for the machines. For example, shift operations can indicate that machine **108A** will complete an excavation task from 2:00 PM-4:00 PM. In another example, shift operations **412** can include table of all machines **108** at the worksite **210** and locations of the machines **108** throughout a work day. The system processor **114** can update the shift operations **412** based on a status of the machines **108**. For example, a machine **108A** can complete a task sooner than expected and notify a worksite controller **112** of the completion. The worksite controller **112** can transmit a signal to the system processor **114** to update a corresponding shift operation **412** for the machine **108A** in memory **118**.

[0054] The charge/discharge controller **116** can receive the schedule from the charging scheduler **404**. For example, a charging scheduler **404** can generate a schedule based on the one or more shift operations **412**. The charging scheduler **404** can transmit the schedule to a charge/discharge controller **116**. The schedule can indicate one or more gaps in operation of the machines **108**. The one or more gaps can be a period of time in which one or more machines **108** are not in use at the worksite **210**. For example, a gap for a machine **108A** can be from 11:00 AM-12:30 PM, whereas a gap for a machine **108B** can be from 9:30 AM-2:00 PM. The load calculator **406** can use the one or more gaps for the generation of the charging rate. For example, a load calculator **406** can calculate a higher charging rate during one or more small gaps for a machine **108A**. In another example, a load calculator **406** can calculate a lower charging rate during one or more larger gaps for a machine **108A**.

[0055] FIG. 5 depicts an example flow diagram **500** of communication for power quality support for the worksite **210** and microgrid **202**. The flow diagram **500** can include all above defined components to facilitate worksite **210** dynamic charging. The microgrid controller **110** and the worksite controller **112** are configured to provide power quality support at the worksite **210**. The power quality support enables the worksite **210** to dynamically charge/discharge the machines **108** based upon the changing layout **300** of the worksite **210**. Thus, improving the productivity and efficiency at the worksite **210**.

[0056] The load predictor **402** can transmit a first data signal including a load prediction to the microgrid controller **110**. For ease of description, data signals can be electrical signals (e.g., analog signals, digital signals), wireless signals (e.g., radio waves), Serial Peripheral Interface (SPI) signals, Universal Serial Bus (USB) signals, among others. At the same time (or substantially the same/in parallel to), the microgrid assets **410** can transmit a second data signal including power, charge, voltage, or frequency information, among others, to the microgrid controller **110**. The microgrid controller **110** can generate metrics (e.g., efficiency metric, load metrics, etc.) based on the information received from the microgrid assets **410**. The microgrid controller **110** can transmit a third data signal including the metrics to the worksite controller **112**. The worksite controller **112**

can transmit a fourth data signal including the metrics to the load calculator **406**. The load calculator **406** may leverage information from the charging scheduler **404** and the metrics to generate a charge requirement. A fifth data signal including the charge requirement can be transmitted from the system processor **114** to the charge/discharge controller **116**.

[0057] The charge/discharge controller **116** can issue a charge command or a discharger command to the machines **108** and the chargers **106**. In some arrangements the machines **108** can send battery metrics to the chargers **106** using Direct Current (DC). In some arrangements, the chargers **106** can send a charger availability to the machines **108** using DC power. The microgrid assets **410** can be sent to the chargers **106** to replenish the power of the chargers **106** during the workday using Alternating Current (AC) power. The chargers **106** can transmit the demand for power to the microgrid assets **410** of the microgrid **202** using AC power.

[0058] FIG. **6** depicts an example flow diagram **600** of communication for economic optimization for the worksite **210** and microgrid **202**. The flow diagram **600** can include all above defined components to facilitate worksite **210** dynamic charging. The system processor **114** can include a weather predictor **408**. The weather predictor **408** can predict weather forecast weather predictions near or at the worksite **210**. In some embodiments, the weather predictor **408** may be or include a third-party resource, such as third-party weather application or forecasting resource. For example, a weather predictor can predict rainy/overcast/sunny weather at a worksite **210** on a particular day/date/range of time. The weather predictor **408** can transmit the predicted weather to the microgrid controller **110**. For example, a weather predictor **408** can notify a microgrid controller **110** of rainy/overcast/sunny weather at a worksite **210**. The microgrid controller **110** can adjust the load metrics based on the predicted weather (e.g., harsh weather, may require a higher load). For example, rainy weather can create mud at a worksite **210**. The mud can create conditions difficult for machines **108** to navigate the worksite **210**. Therefore, a microgrid controller **110** can increase load metrics accordingly. As another example, the microgrid controller **110** can adjust estimates for renewables **208** based on information received from the weather predictor **408**. For instance, the microgrid controller **110** can increase an estimated output of power from renewables **208**, based on a predicted sunny or windy day. The microgrid controller **110** and the worksite controller **112** are configured to provide economic optimization at the worksite **210** by adjusting load metrics and metrics according to the weather while maintaining power support quality. The economic optimization enables the worksite controller **112** to dynamically charge the machines **108** based upon the changing layout **300** and weather of the worksite **210**. Thus, the systems and methods described herein may improve the productivity and efficiency at the worksite **210**.

[0059] The load predictor **402** can transmit a first data signal including a load prediction to the microgrid controller **110**. At the same time, the microgrid assets **410** can transmit a second data signal including power and current efficiency to the microgrid controller **110**. The microgrid controller **110** can generate metrics (e.g., efficiency metric, load metrics, etc.) based on the load prediction, weather prediction, and the current efficiency of the worksite **210** and transmit a third data signal including the metrics to the worksite controller **112**. The worksite controller **112** can transmit a fourth data signal including the metrics to the load calculator **406**. The load calculator **406** may receive a fifth data signal to leverage a charge schedule from the charging scheduler **404** and the metrics to generate a charge requirement. The charge schedule can incorporate shift operations **412** of the one or more machines **108**. The load calculator **406** can receive a sixth data signal including battery information (e.g., state of health, state of charge, voltage, etc.) from the machines **108**. The charge requirement can be transmitted from the system processor **114** to the charge/discharge controller **116**.

[0060] The charge/discharge controller **116** can issue a charge command, within a seventh data signal, to the chargers **106**. In some arrangements the machines **108** can send battery metrics, using DC power, to the chargers **106**. In some arrangements, the chargers **106** can send a charger availability, using DC power, to the machines **108**. The microgrid assets **410** can be sent to the

chargers **106**, using AC power, to replenish the power of the chargers **106** during the workday. The chargers **106** can transmit the demand for power, using AC power, to the microgrid assets **410** of the microgrid **202**.

[0061] FIG. 7 depicts a **700** method to implement worksite dynamic charging for the worksite and microgrid. The method **700** can be performed by, using, or for system **100**. The method **700** can include receiving information indicative of load metrics relating to a power grid at step **705**. The method **700** can include identifying a metric for a plurality of charging machines to be charged at the worksite at step **710**. The method **700** can include determining a charging rate for a charging machine at the worksite at step **715**. The method **700** can include controlling a plurality of chargers at the worksite at step **720**.

[0062] At step **705**, the method **700** can include receiving, by one or more processors (e.g., system processor **114**, microgrid controller **110**, worksite controller **112**, charge/discharge controller **116**) information indicative of load metrics relating to a power grid (e.g., microgrid **202**). The information indicative of the load metrics is received from the plurality of chargers (e.g., chargers **106A-N**) and from a grid controller (microgrid controller **110**) associated with the power grid. The load metrics can include one or more measurements corresponding to a demand in power at the worksite (e.g., worksite **210**), wherein the demand indicating an increase in power at the worksite or a decrease in power at the worksite. For example, the worksite controller **112** can receive information indicative of load metrics from the system processor **114** to determine a charging rate for one or more machines.

[0063] At step **710**, the method **700** can include identifying, by the one or more processors, a metric for a plurality of charging machines (e.g., machines **108A-N**) to be charged at the worksite. The metric for the plurality of charging machines can include one or more measurements corresponding to a supply of charge with the corresponding charging machine. For example, a machine **108A** can have a supply of charge of 20%. In another example, a machine **108B** can have a supply of charge of 90%. The metric can be a value corresponding to an input for a load calculator (e.g., load calculator **406**).

[0064] At step **715**, the method **700** can include determining, by the one or more processors, for each charging machine of the plurality of charging machines, a charging rate for the charging machine at the worksite, according to the metrics and the load metrics of the power grid. The power grid can be a microgrid (e.g., microgrid **202**). The method **700** can include receiving, by the one or more processors, battery metrics of the charging machine. The battery metrics can indicate at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters. For example, the worksite controller can receive the battery metrics from the machines to provide to the load calculator. The method **700** can include determining, by the worksite controller, the charging rate for the charging machine at the worksite, based on a renewable availability (e.g., renewables **208**) from the microgrid, wherein the renewable availability including at least one of wind turbines, photovoltaic (PV) panels, generator sets (e.g., gensets **206**), or hydropower. The method **700** can include determining, by the worksite controller, the charging rate for the charging machine at the worksite, based on an efficiency metric for the microgrid to maximize the efficiency of the microgrid. The charging rate can include a negative charging rate. The method **700** can include discharging the corresponding charging machine to the power grid. For example, the machine can discharge at an operation location (e.g., operation location **104A-N**) to reduce idle time of the machine and correspond to the negative charging rate.

[0065] At step **720**, the method **700** can include controlling a plurality of chargers at the worksite. The method **700** can include receiving, by the one or more processors, a schedule, wherein the schedule indicating one or more gaps in operation of the charging machines. The method **700** can include transmitting, by the one or more processors of a worksite controller, one or more signals to a charge/discharge controller, to cause the charge/discharge controller to transmit one or more signals to the plurality of charging machines. The method **700** can include transmitting, by the

charge/discharge controller, one or more signals to the plurality of chargers. The one or more signals indicating to charge the charging machine or discharge the charging machine. The charge/discharge controller can use the one or more signals to reduce an idle time of the machines. Thus, improving the productivity and efficiency of the worksite.

INDUSTRIAL APPLICABILITY

[0066] The disclosed embodiments may be applicable to any dynamic charging based system or solution. For example, the disclosed embodiments may be applicable or applied to a machine, such as a bulldozer, a forklift, or any other type of machinery, a generator, a transformer, or any other type of electrical utility to be dynamically charged. The disclosed embodiments may be applicable to worksites which include various types of heavy machinery designed to execute tasks throughout the worksite to maximize efficiency. The disclosed worksite controller **112** described herein, may be provided to increase efficiency by generating proper charge and discharge rates for the heavy machinery. For example, because the worksite controller **112** receives information indicating charger availability and states of charge regarding the machines **108** and chargers **106**, the worksite controller **112** can determine a charge/discharge rate for each respective machine. The charge/discharge rate may be dependent upon a charging schedule, state of charge of the machine **108**, or microgrid assets **410**, among others.

[0067] In various embodiments of the present solution, the worksite controller **112** may communicate with a microgrid controller **110** and a charge/discharge controller **116** to continuously maintain a peak efficiency at the worksite **210**. By transmitting information regarding microgrid assets **410**, power, voltage, frequency, charging schedules, and loads, from the various components of the system **100** (e.g., chargers **106**, machines **108**, microgrid controller **110**, charge/discharge controller **116**), the worksite controller **112** can continuously communicate with the machines **108** to maintain a peak efficiency at the worksite **210** and reduce an idle time of the machines **108**.

[0068] In various embodiments of the present solution, the worksite controller **112** may leverage the various components of the system processor **114** to increase productivity at the worksite **210**. For example, the system processor **114** can provide the worksite controller **112** with a plurality of load metrics corresponding to a load at a first machine **108A**. By using the load metrics for the first machine **108A**, the worksite controller **112** can provide a charging rate for the machine **108A** based on changes of the load metrics while the machine **108A** is executing task at the worksite **210**. Furthermore, the worksite controller **112** can trigger the discharge/charge controller to discharge the machine **108A** until a specific time period so the machine **108A** can charge most effectively.

Claims

1. A method, comprising: receiving, by one or more processors, information indicative of load metrics relating to a power grid; identifying, by the one or more processors, metrics for a plurality of charging machines to be charged at a worksite; determining, by the one or more processors, for each charging machine of the plurality of charging machines, a charging rate for the charging machine at the worksite, according to the metrics and the load metrics of the power grid; and controlling, by the one or more processors, a plurality of chargers at the worksite, according to the charging rate associated with a corresponding charging machine.
2. The method of claim 1, wherein the information indicative of the load metrics is received from the plurality of chargers and from a grid controller associated with the power grid.
3. The method of claim 1, wherein the load metrics comprise one or more measurements corresponding to a demand in power at the worksite, wherein the demand indicating an increase in power at the worksite or a decrease in power at the worksite.
4. The method of claim 1, wherein the charging rate for the charging machine comprises a negative charging rate, and wherein the controlling the charger according to the negative charging rate comprises discharging the corresponding charging machine to the power grid.

5. The method of claim 1, wherein the metric for the plurality of charging machines comprises one or more measurements corresponding to a supply of charge with the corresponding charging machine.
6. The method of claim 1, wherein the power grid comprises a microgrid, wherein the one or more processors are of a worksite controller, and wherein controlling the plurality of chargers at the worksite comprises transmitting, by the one or more processors of the worksite controller, one or more signals to a charge/discharge controller, to cause the charge/discharge controller to transmit one or more signals to the plurality of charging machines.
7. The method of claim 6, further comprising transmitting, by the charge/discharge controller, one or more signals to the plurality of chargers, wherein the one or more signals indicating to charge the charging machine or discharge the charging machine.
8. The method of claim 6, further comprising: determining, by the worksite controller, the charging rate for the charging machine at the worksite, based on a renewable availability from the microgrid, wherein the renewable availability including at least one of wind turbines, photovoltaic (PV) panels, generator sets, or hydropower; and determining, by the worksite controller, the charging rate for the charging machine at the worksite, based on an efficiency metric for the microgrid to maximize the efficiency of the microgrid.
9. The method of claim 1, wherein determining a charging rate for the charging machine at the worksite further comprising receiving, by the one or more processors, battery metrics of the charging machine, wherein the battery metrics indicating at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters.
10. The method of claim 1, wherein controlling a plurality of chargers at the worksite further comprises receiving, by the one or more processors, a schedule, wherein the schedule indicating one or more gaps in operation of the charging machines.
11. A data processing system comprising: memory; one or more processors configured to: receive information indicative of load metrics relating to a power grid; identify metrics for a plurality of charging machines to be charged at a worksite; determine for each charging machine of the plurality of charging machines, a charging rate for the charging machine at the worksite, according to the metrics and the load metrics of the power grid; and control a plurality of chargers at the worksite, according to the charging rate associated with a corresponding charging machine.
12. The data processing system of claim 11, wherein the information indicative of the load metrics is received from the plurality of chargers and from a grid controller associated with the power grid.
13. The data processing system of claim 11, wherein the load metrics comprise one or more measurements corresponding to a demand in power at the worksite, wherein the demand indicating an increase in power at the worksite or a decrease in power at the worksite.
14. The data processing system of claim 11, wherein the charging rate for the charging machine comprises a negative charging rate, and wherein, to control the charger according to the negative charging rate, the one or more processors are configured to discharge the corresponding charging machine to the power grid.
15. The data processing system of claim 11, wherein the metrics for the plurality of charging machines comprise one or more measurements corresponding to a supply of charge with the corresponding charging machine.
16. The data processing system of claim 11, wherein the grid comprises a microgrid, wherein the one or more processors are of a worksite controller, and wherein when controlling the plurality of chargers at the worksite, the one or more processors are configured to transmit one or more signals to a charge/discharge controller, to cause the charge/discharge controller to transmit one or more signals to the plurality of charging machines.
17. The data processing system of claim 16, wherein the charge/discharge controller further configured to transmit one or more signals to the plurality of chargers, wherein the one or more signals indicating to charge the charging machine or discharge the charging machine.

18. The data processing system of claim 16, wherein the one or more processors are further configured to: determine the charging rate for the charging machine at the worksite, based on a renewable resource availability from the microgrid, wherein the renewable resource availability includes at least one of wind turbines, photovoltaic (PV) panels, generator sets, or hydropower; and determine the charging rate for the charging machine at the worksite, based on an efficiency metric for the microgrid to maximize efficiency of the microgrid.

19. The data processing system of claim 11, wherein when determining a charging rate for the charging machine at the worksite, the one or more processors are configured to receive battery metrics of the charging machine, wherein the battery metrics indicate at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters.

20. The data processing system of claim 11, wherein when controlling the plurality of chargers at the worksite, the one or more processors are further configured to receive a schedule, wherein the schedule indicates one or more gaps in operation of the charging machines, wherein the gap is a period of time in which the charging machine is not in use at the worksite.
